

Algebra & Polynomials

Marking Scheme & Blueprint

MTH1W · Module 2 Test — Teacher Copy

Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

Q	Sub-topic	Level	Marks	Category	Section
A1	Like Terms	L1	1	K/U	A
A2	Like Terms	L1	1	K/U	A
A3	Substitution	L1	1	K/U	A
A4	Add Polynomials	L1	1	K/U	A
A5	Multiply by Constant	L1	1	K/U	A
A6	Divide by Constant	L1	1	K/U	A
A7	Exponent Laws	L1	1	K/U	A
A8	Like Terms	L1	1	K/U	A
A9	Polynomials	L1	1	K/U	A
A10	Like Terms	L1	2	K/U	A
A11	Substitution	L1	2	K/U	A
A12	Subtract Polynomials	L1	2	K/U	A
A13	Multiply by Monomial	L1	2	K/U	A
A14	Divide by Monomial	L1	2	K/U	A
A15	Exponent Laws	L1	2	K/U	A
B16	Like Terms	L2	2	App	B
B17	Add Polynomials	L2	2	App	B
B18	Multiply by Constant	L2	2	App	B
B19	Divide by Constant	L2	2	App	B
B20	Exponent Laws	L2	2	App	B
B21	Substitution	L2	2	App	B
B22	Multiply/Expand	L2	3	App	B
B23	Subtract Polynomials	L2	3	App	B
B24	Divide by Monomial	L2	3	App	B
C25	Perimeter	L3	4	Think	C
C26	Expand & Collect	L3	4	Think	C
C27	Area & Perimeter	L3	5	Think	C
C28	Formula & Evaluate	L3	5	Think	C
C29	Multi-technique	L3	6	Think	C
D30	Like Terms vs Product	L3	4	Comm	D

By category (marks): K/U 21 · Application 21 · Thinking 24 · Communication 4

By level (marks): L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**
= 5x. (A1)
- 2. (A2 · K/U · 1 marks)**
= 2a + 7. (A1)
- 3. (A3 · K/U · 1 marks)**
 $4(2)+1 = 9$. (A1)
- 4. (A4 · K/U · 1 marks)**
= 5x + 3. (A1)
- 5. (A5 · K/U · 1 marks)**
= 5x - 15. (A1)
- 6. (A6 · K/U · 1 marks)**
= 2x + 3. (A1)
- 7. (A7 · K/U · 1 marks)**
 $x^{4+5} = x^9$. (A1)
- 8. (A8 · K/U · 1 marks)**
-6. (A1)
- 9. (A9 · K/U · 1 marks)**
3 terms. (A1)
- 10. (A10 · K/U · 2 marks)**
 $3m + 5n$. (M1, A1)
- 11. (A11 · K/U · 2 marks)**
 $6 + (-3) = 3$. (M1, A1)
- 12. (A12 · K/U · 2 marks)**
 $5x - 2x + 4 + 3 = 3x + 7$. (M1, A1)
- 13. (A13 · K/U · 2 marks)**
 $6x^2 + 10x$. (M1, A1)
- 14. (A14 · K/U · 2 marks)**
 $3x - 2$. (M1, A1)
- 15. (A15 · K/U · 2 marks)**
 $8x^5$. (M1 coeff, A1)
- 16. (B16 · Application · 2 marks)**
 $(6x^2+3x^2) + (-2x-5x) = 9x^2 - 7x$. (M1, A1)

- 17. (B17 • Application • 2 marks)**
 $4a^2 + 3a - 4$. (M1, A1)
- 18. (B18 • Application • 2 marks)**
 $-6x + 12$. (M1, A1)
- 19. (B19 • Application • 2 marks)**
 $2x^2 + x$. (M1, A1)
- 20. (B20 • Application • 2 marks)**
 Coefficients $20 \div 5 = 4$; subtract exponents: $4x^4$. (M1, A1)
- 21. (B21 • Application • 2 marks)**
 $16 - 8 + 1 = 9$. (M1, A1)
- 22. (B22 • Application • 3 marks)**
 $4x + 8 + 6x - 3 = 10x + 5$. (M1, M1, A1)
- 23. (B23 • Application • 3 marks)**
 $7x^2 - 2x^2 = 5x^2$; $3x + 5x = 8x$; $-2 - 4 = -6$.
 $= 5x^2 + 8x - 6$. (M1, M1, A1)
- 24. (B24 • Application • 3 marks)**
 Divide each term by $4x$: $3x^2 - 2x + 1$. (M1, M1, A1)
- 25. (C25 • Thinking • 4 marks)**
 $\text{Third} = (12x + 3) - (4x - 1) - (5x + 5)$. (M1)
 $= 12x + 3 - 4x + 1 - 5x - 5$. (M1)
 $= 3x - 1$. (A1, A1)
- 26. (C26 • Thinking • 4 marks)**
 $2x(3x - 4) = 6x^2 - 8x$; $3(x^2 - 2x) = 3x^2 - 6x$. (M1, M1)
 $6x^2 - 8x - 3x^2 + 6x = 3x^2 - 2x$. (A1, A1)
- 27. (C27 • Thinking • 5 marks)**
 $\text{Length} = (6x^2 + 9x) \div 3x = 2x + 3$. (M1, A1)
 $\text{Perimeter} = 2[(2x + 3) + 3x] = 2(5x + 3)$. (M1)
 $= 10x + 6$. (A1, A1)
- 28. (C28 • Thinking • 5 marks)**
 $P = 2(3x + 1) + 2(x + 4) = 6x + 2 + 2x + 8$. (M1, M1)
 $P = 8x + 10$. (A1)
 At $x = 5$: $8(5) + 10 = 50$. (M1, A1)
- 29. (C29 • Thinking • 6 marks)**
 $(10x^3 - 15x^2) \div 5x = 2x^2 - 3x$. (M1)
 $2x(4 - x) = 8x - 2x^2$. (M1)
 $2x^2 - 3x + 8x - 2x^2 - 3$. (M1, M1)
 $= 5x - 3$. (A1, A1)

30. (D30 • Communication • 4 marks)

Adding: $3x + 3x = 6x$ (add coefficients, keep the variable). (A1)

Multiplying: $3x \times 3x = 9x^2$ (multiply coefficients, add exponents). (A1)

So Aisha is not correct; $3x + 3x = 6x$. (C1, C1)